

SYSTEMATIC REVIEW

Open Access



Barriers and facilitators for implementation of lifestyle related blended care interventions for overweight, obesity and type 2 diabetes mellitus: a scoping review

Chiara Lansink^{1,2*} , Anouk Middelweerd¹, Gozewijn Dirk Laverman^{1,2} and Nienke Beerlage-de Jong³

Abstract

Background Blended care, referring to the combination of digital health interventions (DHI) with in-person care, has proven cost-effective in promoting healthy behaviours, especially for lifestyle related conditions like type 2 diabetes mellitus. However, despite positive results, many interventions fail during the clinical implementation. A successful implementation of blended care interventions requires a systematic approach including appropriate implementation strategies that address relevant barriers and facilitators. Therefore, this study explores barriers and facilitators for implementation of lifestyle related blended care interventions for weight management and type 2 diabetes mellitus.

Methods Following PRISMA-ScR guidelines, a scoping review was conducted using PubMed, Scopus, and Web of Science, covering studies from January 1, 2000, to February 4, 2025. The search included terms related to 'overweight', 'obesity', 'diabetes mellitus type 2', 'eHealth', 'blended care', 'setting' and 'implementation'. Studies were included if they focused on lifestyle changes (e.g. weight loss, physical activity, or self-management) and reported barriers or facilitators for implementing blended care, defined in this review as the integration of DHI with at least one in-person care-focused interaction between a health care provider and an adult patient. No risk of bias assessment was performed. Data were extracted on study characteristics, participants demographics, DHI delivery methods, and implementation barriers and facilitators. Data was analysed using a narrative synthesis structured by the Consolidated Framework for Implementation Research (CFIR).

Results From 1329 screened studies, nine met the inclusion criteria. The findings indicate that implementation challenges cluster around financial constraints, increased workload for healthcare providers, and disparities in digital skills, which collectively limit the feasibility and sustained use of blended DHIs. The other way around, successful implementation is supported by adequate training, user-centred design, and strong alignment with existing clinical workflows. Tailoring interventions to organisational context and ensuring regulatory compliance further strengthen adoption.

Conclusions Blended DHIs offer opportunities to enhance personalization, patient involvement, and continuity of care. However, their sustainable implementation requires addressing these overarching barrier and facilitator domains rather than isolated factors: Healthcare organisations can act to reduce organisational burden by providing sufficient resources, policymakers can support regulatory alignment, developers can prioritize user-centred design, and healthcare professionals can integrate blended care into routine practice and support patients in its use.

*Correspondence:

Chiara Lansink

c.i.lansink@utwente.nl

Full list of author information is available at the end of the article



© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

Keywords Implementation science, Blended care, Digital health, EHealth

Contributions to the literature

- By structuring findings using the Consolidated Framework for Implementation Research (CFIR), This study provides a comprehensive perspective on the implementation of blended care interventions.
- Unlike already existing studies focusing more on the effectiveness of blended care, this study contributes by identifying specific barriers and facilitators for its implementation. Identified barriers include high costs, integration issues and challenges related to user-friendliness, while factors include the application of user-centred design approaches and personalized care strategies.
- This study identifies the need for consistent terminology and clear definitions of blended care to improve comparability, knowledge transfer and allow for cross comparisons.

Introduction

Diabetes Mellitus type 2 (T2DM) is one of the four major non-communicable diseases [1]. In 2021, the global prevalence of diabetes among individuals aged 20 to 79 reached 536.6 million and is expected to increase to 783.2 million by 2045 [2, 3]. Individuals with overweight and obesity are at a higher risk of developing type 2 diabetes [4]. As the number of people living with overweight and obesity continues to grow, so too are associated healthcare costs [5].

Lifestyle factors, like diet and physical activity, are key in managing and even reversing T2DM [6, 7]. However, adopting and maintaining a healthy lifestyle is challenging for many individuals, partly because of different levels of motivation and self-management skills [8, 9]. For people with T2DM, these challenges are further compounded by the broader demands of self-management, which includes not only making lifestyle changes but also adhering to medication and monitoring blood glucose levels [10–12]. Effective self-management helps individuals take control of their health, manage symptoms, and improve their quality of life [13–15]. Yet, despite clinical guidelines emphasising lifestyle change as the first step in treatment, existing programs, such as combined lifestyle interventions, often lack sufficient support to facilitate sustainable lifestyle changes [16].

The rapid growth of interactive digital health interventions (DHI), including smartphone apps for tracking diet and physical activity, has proven effective in

promoting healthy behaviours and achieving weight loss goals [17, 18]. These innovative, cost-effective tools may enhance self-management and help reduce healthcare costs [11, 13, 19, 20]. However, stand-alone DHI may be less effective compared to a combination of digital approaches with at least one in-person visit [11, 21–25]. Such blended care, which integrates eHealth with in-person interactions, combines the benefits of digital tools with traditional care, potentially improving healthcare quality, efficiency and patient satisfaction [13, 25, 26]. While several systematic reviews have found promising outcomes of digital and eHealth interventions targeting physical activity, healthy diet or improving self-management in adults [27–31], studies that explicitly examine blended care remain scarce [25]. The few reviews that include blended elements typically do so as part of broader eHealth or lifestyle interventions, rather than evaluating blended care as a distinct approach [32, 33]. For this study we defined blended care as a digital health intervention combined with at least one in-person instance between health care provider and patient.

To optimize the effect of blended care, effective implementation of DHI in a blended care setting is essential to fully realize their benefits. However, implementing these innovations in healthcare is difficult and they often do not last in the long term [34]. Research suggests that the success of implementing DHI depends on a combination of organisational, technological, and human factors. These factors function as determinants that may positively or negatively influence implementation outcomes. Challenges such as financial constraints and difficulties integrating interventions into routine care pathways can hinder adoption, reduce patient engagement, and ultimately limit the effectiveness of these interventions [35–37]. Such challenges may reduce uptake and threaten long-term sustainability. Furthermore, implementation strategies are often underreported, limiting their practical application [38]. To better understand these factors, the Consolidated Framework for Implementation Research (CFIR) provides a useful structure and is applied in this review [39]. CFIR provides a comprehensive framework for identifying determinants that influence implementation processes and outcomes. Previous reviews have explored the barriers and facilitators to adopting and implementing DHI among healthcare professionals [40–42]. These reviews shows that the adoption of DHI is mainly influenced by perceived usefulness and ease of use, with limited knowledge of eHealth being a major barrier.

In addition, good alignment with work processes, end-user involvement, and collaboration between healthcare professionals and IT specialists are important factors. Although these reviews provide valuable insights into the broader implementation of digital health, they do not specifically examine how DHIs are combined with in-person care. Consequently, explicit evidence on how blended care is implemented in practice remains limited.

This review aims to bridge this gap by identifying key barriers and facilitators influencing the implementation outcomes of blended care interventions for lifestyle management in adults with overweight, obesity and/or T2DM. By synthesizing the results, the aim is to optimize healthcare delivery, enhance patient outcomes and support sustainable lifestyle changes.

Methods

Protocol and registration

This review has been conducted in accordance with the Preferred Reporting for Systematic Reviews and Meta-Analysis Extension for Scoping Reviews guidelines, using the PRISMA-ScR checklist [43]. No formal protocol was registered for this review.

Eligibility criteria

The inclusion criteria for articles were as follows: (1) the publication must be in English or Dutch; (2) participants must be healthcare professionals directly involved in the care for patients with overweight, obesity and/or T2DM; and/or (3) participants must be adults (>18 years) with T2DM and/or obesity/overweight. Participants may present with a combination of metabolic syndrome criteria or cardiovascular risk factors. Studies were eligible if they included healthcare professionals, adult patients or both; (4) the publication must be a peer-reviewed scientific journal article or conference paper; (5) the intervention must be aimed at lifestyle changes such as weight loss or increasing physical activity or improving self-management; (6) the study must focus on barriers or facilitators for implementing blended care, defined in this review as the integration of a DHI with at least one instance of in-person care between health care provider and patient. The exclusion criteria included the following: (1) participants with co-morbidities that are not related to metabolic syndrome or cardiovascular risk factors; (2) participants with serious mental illness; (3) participants who are pregnant; and (4) participants with substance addiction.

Information sources and search strategy

A search was performed for articles between 1st of January 2000 and the date of search (4th February 2025), on PubMed, Scopus, and Web of Science. These databases

were selected because together they cover the fields most relevant to this review, including public health, technological, and implementation research related to blended care. The key words that were used for the search were 'overweight,' 'obesity,' diabetes mellitus type 2,' eHealth,' 'blended care,' 'setting' and 'implementation.' The complete electronic search strategy for the PubMed database is detailed in Additional file 1. To ensure consistency among all databases, the same search strategy was applied for the three databases, PubMed, Scopus, and Web of Science. All searches were restricted to articles published in English or Dutch. Only journal articles, conference papers, and meta-analyses were included, with all other publication types excluded.

Selection of studies

Duplicates were removed through Covidence [44], and the first author manually removed the remaining duplicates. The first author (CL) screened the titles and abstracts of all publications, while a second author (NB) independently screened 20% of these titles and abstracts. Initial agreement during title and abstract screening was limited (Cohen's kappa, $\kappa=0.29$), reflecting variability in the interpretation of the term 'blended care.' Discrepancies were discussed to clarify the inclusion criteria and improve consistency. A similar procedure was applied to full-text screening, with initial agreement also limited (Cohen's kappa, $\kappa=0.26$) but improved through discussion and refinement of the criteria, ensuring reliability in the final selection of studies. A revised inter-rater agreement was not calculated, as its purpose in this context was to identify discrepancies rather than to quantify final agreement. For this study we defined blended care as a digital health intervention combined with at least one in-person instance between health care provider and patient.

Data charting process

The standard Covidence data extraction template was adapted for the purpose of this scoping review. The adapted extraction form was pilot tested with three studies, based on which the form was modified to its definitive version. The extracted data covered: (1) Study characteristics (e.g. publication type); (2) general study and participants characteristics (e.g. study design, country); (3) mode of intervention delivery intervention if applicable (e.g. delivery platform); (4) qualitative data methods (e.g. interview type); (5) evaluation results (e.g. barriers and facilitators; and (6) other information (e.g. key conclusion of study authors). The data extraction was conducted by CL and then checked for accuracy and completeness by NB. Any discrepancies were resolved through consultation and consensus.

Synthesis of results

In line with the exploratory nature of this review, no formal risk of bias assessment was performed. Given the predominantly qualitative nature of the included studies, findings were synthesized using a narrative approach. For consistency in reporting, the term digital health intervention (DHI) was used throughout this scoping review to specifically refer to the digital component of the blended-care interventions included. To identify the implementation determinants in a structured and systematic way, the Consolidated Framework for Implementation Research (CFIR) was used. Given the complexity of implementing digital health interventions, a framework that systematically addresses contextual, organisational, and individual influences was considered most appropriate. CFIR consisted of five domains that can influence implementation. These domains include innovation (e.g. innovation type and adaptable components), outer setting (e.g. external influences), inner setting (e.g. influences within the setting), individuals (e.g. characteristics and roles of the individuals involved) and implementation (e.g. activities and strategies used to implement the innovation) [39]. This framework provides a systematic method for understanding how different components interact and influence the overall success of DHI implementations. Relevant sentences addressing barriers and facilitators in the included studies were coded deductively according to the five CFIR domains and their constructs. Coding was conducted by CL and discussed with a second reviewer (NB), with any discrepancies or uncertainties resolved through discussion and consensus. No inductive coding was performed, as all relevant factors were captured within the existing CFIR framework.

Results

Search results

The search terms resulted in 2637 articles from three databases. Of these, 1105 (41.9%) were duplicates and were removed. There was initially a relatively low consensus for the first 10% of the data. However, after discussing the discrepancies, particularly those arising from varying interpretations of terms, agreement improved considerably. After screening titles and abstracts, 179 articles were selected for full-text reviews. Of these, 170 articles were excluded, predominantly because they did not have an in-person component. Eventually, nine studies were included in this scoping review (Fig. 1).

Study characteristics

The included studies were conducted across six different countries; Sweden ($n=2$) [45, 46], Spain ($n=2$) [47, 48], United Kingdom ($n=2$) [38, 49], United States ($n=1$) [50], Australia ($n=1$) [51], the Netherlands ($n=1$) [52].

Most studies ($n=4$) focused on both healthcare professionals and patients [45, 48–50], while two focused solely on healthcare professionals [51, 52]. Two studies included a broad range of health professionals, including GPs, nurses, healthcare assistants, administrative staff, and regional healthcare directors [38, 45]. Three studies included individuals at high risk of developing T2DM, based on the Finnish Diabetes Risk Score (FINDRISC) [46–48]. Of these, one study included participants with T2DM, pre-diabetes, and high-risk individuals [46], another focused on primary care professionals and high-risk patients [48], and the third included only high-risk participants [47].

The DHIs evaluated across these studies varied in focus and implementation. Two studies [47, 48] assessed the same DHI, the DE-PLAN program, implemented in different Spanish regions; this intervention centred on education and structured contact between healthcare professionals and patients. Another study [52] evaluated a DHI offering individualised coaching and remote monitoring through telephone consultations. Two studies [46, 50] focused on communication-enhancing DHIs such as video consultations to facilitate patient-provider interaction. One study [49] investigated a remote monitoring intervention tracking health indicators, and another [38] explored an education-focused DHI aimed at improving diabetes and lifestyle knowledge among patients. Finally, two studies out of nine in this review used Theories, Models or Frameworks (TMF) to guide their implementation processes; Reach, Effectiveness, Adoption, Implementation, Maintenance (RE-AIM) [47] and Normalization Process Theory (NPT) [38].

A summary of the study characteristics is presented in Table 1, while the full table can be found in Supplementary Table 1, Additional file 2.

A total of six studies [45–50] described patients' perspective on the implementation of blended care for lifestyle interventions in the area of T2DM, overweight/obesity or high risk of developing T2DM. Additionally, 8 studies [38, 46–51] described the perspective of the healthcare professional. All studies were conducted within a primary care setting and primarily highlighted barriers rather than facilitators to successful implementation. Seven studies [38, 46–50, 52] focused specifically on barriers to implementing a specific intervention while two studies [45, 51] focused on barriers and facilitators more generally in the development of a blended care intervention. While, studies referred to telemedicine [50], web-based interventions [38, 52], virtual health [47, 48], telemonitoring [49], eHealth services [45], telephone facilitated intervention [46] or mobile health applications [51]. Although none of the studies identified the intervention as blended care, all

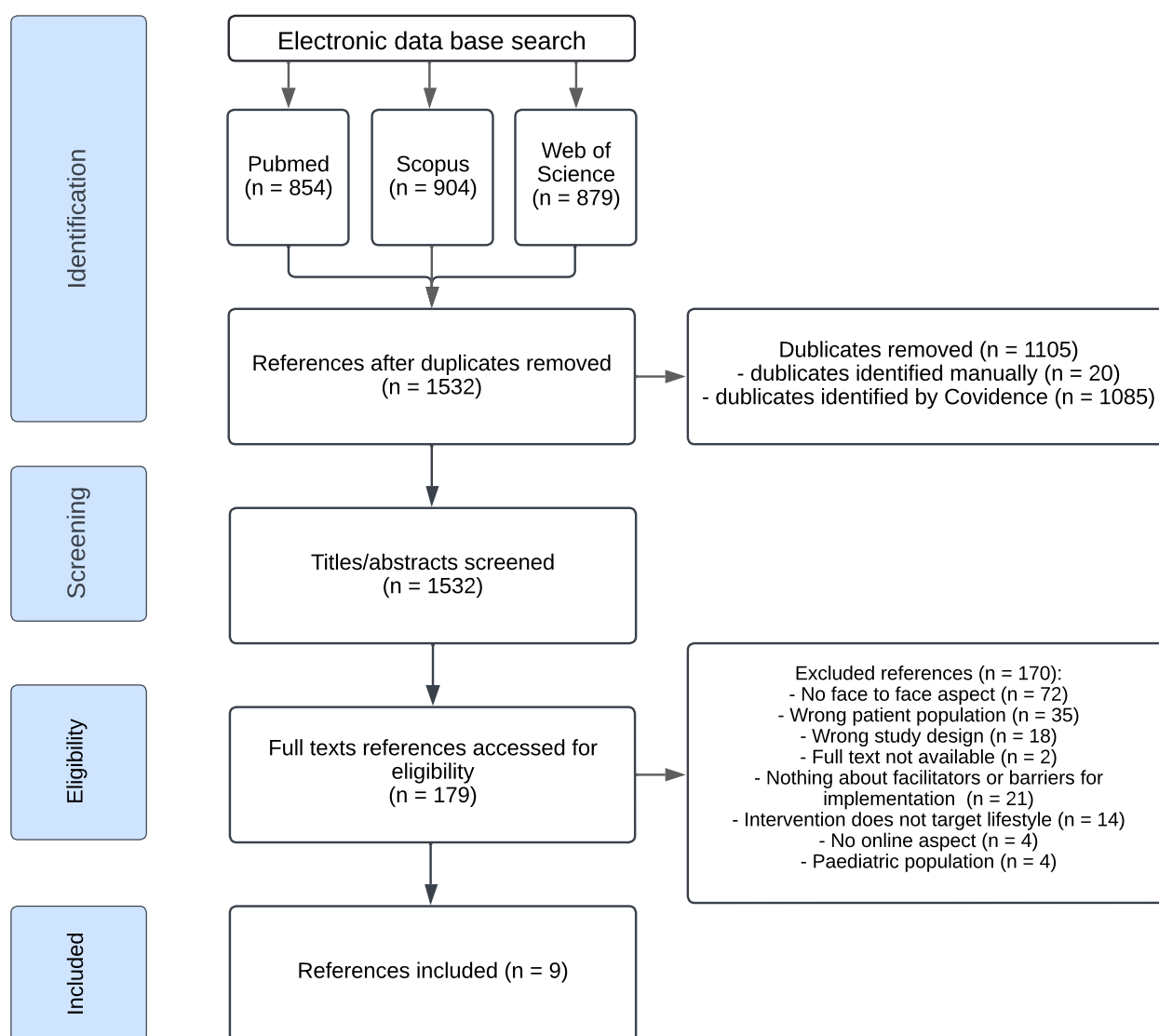


Fig. 1 PRISMA flowchart illustrates stages of the literature search

studies implemented a DHI alongside in-person care. Among the studies focusing on a specific intervention, some combined the DHI with usual care [38, 47, 49, 52], while others [46, 48, 50] incorporated group sessions; including 2 sessions [46], 16 sessions [48], or 8 sessions [50]. Two [45, 51] additional studies focused on general barriers and facilitators of blended care rather than a specific intervention.

Consolidated framework for implementation research

The results of this review are summarized according to the domains of the consolidated framework for implementation research (CFIR) [39]. Additional file 3 provides a table for easy understanding of the identified factors for

implementing DHI in blended healthcare using the CFIR framework.

Innovation domain

Several studies have identified key requirements within the innovation domain that can contribute to successful adoption and integration in healthcare [45, 46, 49, 51, 52]. In terms of relative advantage, GPs generally recognized that DHIs can enhance personalized care [45, 46, 51, 52] and support patient safety between consultations. Features like notifications and reminders were seen as having the potential to encourage more active patients' engagement [51]. Other values attributes included visual orientation, ease of use, and the constant availability for

Table 1 Summary of the characteristics of the included studies

Author and year (ref)	Country	Study design	Participants	Target	eHealth tool	Frequency face to face care	Theories, Models or Frameworks (TMF) used
Ayre, 2019 [51]	Australia	Qualitative (semi-structured interviews)	25 primary care physicians	Self-management (medication adherence, physical activity, healthy diet and weight management)	Mobile health application	N/A	N/A
Brown, 2020 [50]	United States	Pre-post intervention study	86 obesity patients & 7 clinics	Weight loss	Telemedicine	8 group sessions	N/A
Costa-Pinel, 2018 [48]	Spain	Mixed method	1458 patients of high risk of developing diabetes based on FINDRISC > 11 and/or pre-diabetes diagnosis & 305 primary care professionals	Lifestyle, physical activity and nutritional intervention	Virtual health program	16 group sessions	N/A
Hanley, 2015 [49]	United Kingdom	Qualitative (semi-structured interviews)	23 T2DM patients & 10 primary care professionals (4 GPs and 6 nurses)	Physical activity, reducing salt in the diet and weight control	Telemonitoring	Usual care	N/A
Ross, 2018 [38]	United Kingdom	Qualitative (semi-structured interviews)	21 health professionals (GPs, practice nurses, diabetes specialist nurses, health care assistants, administrative and reception staff, practice managers and commissioners of care)	Medical management (e.g. eating healthily, exercising more, and stopping or cutting down smoking and drinking alcohol), education, lifestyle changes, emotional management, and social support	Web-based intervention	Usual care	Normalization process Theory (NPT)
Sánchez, 2016 [47]	Spain	Observational descriptive process study	454 patients of high risk of developing T2DM based on FINDRISC score ≥ 14) and 634 control patients	Lifestyle, physical activity and nutritional intervention	Virtual health program	Usual care	Reach, Effectiveness, Adoption, Implementation, Maintenance (RE-AIM)
Schimmer, 2019 [45]	Sweden	User-centred design approach	26 participants (patients, spouses, diabetes specialist nurses, physician and regional primary care director, researchers and system developers)	Self-management (physical activity and diet)	E-health service	N/A	N/A
Timm, 2021 [46]	Sweden	Explanatory sequential mixed methods	Facilitators and 72 T2DM, pre-diabetes or at high-risk of developing diabetes based on (> 13/25 FINDRISC) patients	Self-management (lifestyle behaviour, physical activity, and diet)	Telephone facilitated intervention	2 group sessions	N/A

Table 1 (continued)

Author and year (ref)	Country	Study design	Participants	Target	eHealth tool	Frequency face to face care	Theories, Models or Frameworks (TMF) used
Vonken, 2023 [52]	Netherlands	Qualitative (semi-structured interviews)	16 GPs	Self-management (physical activity, unhealthy snack intake, medication adherence, and smoking)	Web-based intervention	Usual care	N/A

N/A Not Applicable

home monitoring [52]. Adaptability emerged as another construct, with integration into existing healthcare systems emphasized as important for practical implementation [51, 52]. In contrast, complexity was identified as a barrier. Reported challenges included complicated interfaces [45] and language related challenges in understanding educational content and participating in intervention meetings, sometimes requiring support from family members [46]. However, despite concerns about complexity, DHIs were largely viewed as a valuable complement to in-person care [51]. Finally, trialability functioned as a facilitator. Both patients [49] and healthcare professionals [49, 52] valued the opportunity to test DHIs before committing long-term, highlighting that perceived effectiveness, user experience, and patient involvement all play central roles in adoption decisions [52].

Outer setting domain

Although the outer setting was not frequently mentioned in the studies [51, 52], several themes suggest that external factors influence implementation of DHIs. Several outer setting barriers emerged. Liability concerns and perceived obligations to respond promptly to patient data were mentioned [51], reflecting the moral and legal pressures that influence clinical decision-making. Regulatory requirements, particularly regarding data protection, were also aligning with policies and laws highlighting the broader challenge of aligning digital interventions with legal regulations [52]. Financing related issues further complicated implementation. Uncertainty about reimbursement procedures and the risk of uncompensated work were reported [51]. In contrast, some external factors functioned as facilitators. Support from larger organisations was seen as an important facilitator for adoption, reflecting the relevance of partnerships and connections. Additionally, lack of reimbursement was not perceived as a limiting factor by some professionals [52], suggesting that financial barriers may be context dependent rather than universally limiting.

Inner setting domain

Healthcare professionals encountered a variety of challenges from the inner setting, when using DHI for patients [38, 45, 48–52]. Several barriers were identified. A major concern was that a patient monitoring app might create a false sense of security, as patients might think their doctor is constantly checking their data [51]. Professionals perceived that DHIs could either challenge the doctor–patient relationship. Although some professionals indicated that DHI could improve communication [52], the majority believed that face-to-face care remained essential for tailoring information to the patient and ensuring understanding [51, 52]. Workload

was another important factor. Increased workload was described as a barrier to implementation [48, 49]. However, under certain conditions workload was perceived as acceptable, as others considered a temporary rise in workload acceptable during the learning phase [51]. This suggests that workload may be function as a facilitator when it supports adaptation and learning, but becomes a barrier when the increase is substantial or prolonged. To minimize this barrier and enhance compatibility, it is therefore important that DHI be integrated with existing workflows, systems, and processes [38, 51, 52].

In addition, available resources barriers were reported. Concerns about adoption costs [49, 52], were intertwined with practical issues such as limited staff, insufficient space and unstable internet connections [50]. Implementation was further complicated by role uncertainty, highlighting that clear definition of staff responsibilities is crucial for smooth adoption and to prevent procedural barriers [38], which align with structural characteristics. Several facilitators were also identified within the inner setting. Access to knowledge and information, particularly though training of healthcare professionals was seen as essential [38] and described in several studies [46–48]. However, in practice, training needs to be relevant and of sufficient duration. Long online modules with limited follow-up acted as a barriers, as they hindered healthcare professionals from fully engaging with the technology [45, 46].

Individuals' domain

The roles and characteristics of individuals can influence the implementation of digital health innovations [47–49, 51, 52]. Adoption was hindered by digital divide related barriers. These occurred at both patient and system levels, including limited digital skills and health literacy as well as technological accessibility, especially for elderly patients [51, 52], reflecting capability constraints. Professionals noted that DHI tended to motivate more already independent patients, while vulnerable patients required additional support [51], indicating that DHI may inadvertently increase inequalities if additional guidance is not provided. Personalized care remained essential, as online options were often perceived as impersonal [52] and less effective in ensuring patient understanding [51]. Patient preferences also influenced engagement, for example, many participants favoured group-based interventions over individual or mixed formats [47]. In contrast, several facilitators were also described. DHI were appreciated for enhancing patient autonomy and supporting remote care [52]. In addition, knowing that healthcare professionals could see their self-monitoring data motivated behaviour change by patients through a sense of supportive oversight [49]. Successful

implementation also depended on the involvement and support of key stakeholders. Healthcare professionals acted as innovation deliverers, while managers and supervisors supported them as implementation leads or mid-level leaders. Active involvement of both groups functioned as a facilitator, whereas lack of involvement posing a major barrier [48]. Highlighting that healthcare professionals.

Implementation process domain

Several studies have reported barriers and facilitators of the implementation process [38, 47, 48, 52]. Two of the included studies applied implementation frameworks to guide the process, namely Reach, Effectiveness, Adoption, Implementation, Maintenance (RE-AIM) [47] and Normalization Process Theory (NPT) [38], reflecting structured approaches to planning and assessing context. Key challenges include discontinuity of initial efforts and limited sustained commitment from professionals, which can impede integration [48]. For example, in the HeLP-Diabetes intervention, professionals recognized the value of the program but struggled to allocate sufficient time and effort for its effective implementation [38], indicating that perceived value alone was insufficient to ensure sustained engagement. In contrast, multiple facilitators were described. Providing healthcare professionals with opportunities to reflect on the intervention's value through patient feedback was perceived to potentially increase their engagement [38], corresponding to reflecting and evaluating implementation. Moreover, involving patients in decisions regarding the use of DHIs can enhance compatibility with user needs and preferences [52], thereby facilitating alignment with the needs of innovation recipients. Finally, early and iterative implementation, preferably before developing the development phase, can promote adoption, allowing end users to be involved at an early stage [38], illustrating the importance of small implementation cycles (doing) and ongoing adapting.

Discussion

This study aimed to provide a comprehensive overview of the barriers and facilitators involved in implementing blended care interventions for lifestyle management in adults with overweight, or T2DM.

It is notable that the term “blended care” was not always explicitly used in the included studies, but all included a combination of a form of DHI with at least one physical appointment between a healthcare professional and an adult patient. This variability in terminology contributed to the initially low inter-rater reliability during study selection, which was resolved through discussion

and consensus. No clear differences were found between countries regarding the implementation of these DHIs. Importantly, the absence of a standardized definition of “blended care” may in itself constitute an implementation barrier. Rather than constituting a direct barrier in clinical practice, the lack of a standardized definition primarily complicates the accumulation and transfer of implementation knowledge. When terminology varies, evidence becomes fragmented, making it more difficult to develop theoretically informed and empirically grounded implementation strategies. This highlights the importance of using consistent terminology and clear definitions when describing DHIs and blended care. Uniform language would improve comparability between studies, support knowledge transfer, and help address shared implementation challenges across different contexts. Moreover, it is notable that all of the included studies were conducted in primary care, and none focused on secondary care. This may reflect the current emphasis on prevention and early intervention for T2DM in primary care but also raises questions about the applicability and integration of DHIs in more specialized or hospital-based care settings. The DHIs examined serve different roles within blended-care settings, including education, coaching, remote monitoring, and communication with healthcare professionals. These varying functions influence patient engagement and the interpretation of outcomes. For example, coaching and monitoring interventions often require active patient involvement and continuous provider support, whereas communication tools may primarily facilitate access and interaction. Recognising these distinct functions is key to tailoring implementation strategies and ensuring effective integration of DHIs into existing care pathways.

Unlike recent reviews focusing more on the effectiveness of DHI [18, 53–56], this study contributes by identifying specific barriers and facilitators for its implementation. However, interestingly, most of the included studies focused on the effectiveness and design features of the interventions, while implementation-related aspects were often not reported. This implies that implementation to date still does not receive the attention it deserves. By structuring findings using CFIR, this study provides a comprehensive perspective on the implementation of blended care interventions. The results were synthesized narratively, which provided a descriptive overview of facilitators and barriers but does not quantify the prevalence of specific factors. After all, early consideration of implementation increases the chances of developing an intervention that can be more widely applied and sustained in practice [57]. In line with earlier implications of the study by Lee [58] and Ahmed [40], results of the current review suggest that successful

adoption of DHI technologies requires active involvement of both patients and healthcare professionals in the development and implementation stages [38]. Importantly, to strengthen the evidence base, future trials and evaluations should prospectively include implementation outcomes to reflect alignment of implementation strategies and adopt standardized reporting frameworks to allow cross-comparisons between studies. This will improve transparency, enable knowledge transfer, and provide actionable insights into the factors that influence implementation.

To support structured thinking on implementation, theories, models and frameworks (TMF) such as the CFIR [39] and Non-adoption, Abandonment, Scale-up, Spread and Sustainability (NASSS) [59], RE-AIM [60] or NPT [61] offer valuable guidance for the implementation process. They provide a structured approach to understanding the factors that influence implementation outcomes [62], including the involvement of end users. The use of CFIR allowed us to capture the complexity inherent to digital health implementation by systematically identifying determinants across contextual, organisational, and individuals levels. However, the current scoping review found that only two studies reported on applying a TMF to guide their implementation. This is in line with a recent review [63] that found that only 23 studies out of 252 studies reported using an implementation science TMF, which underscores the gap between theory and practice. Adopting an iterative approach, guided by frameworks, ensures the strategy is all-encompassing and can adapt to emerging needs and challenges, ensuring a more effective and responsive implementation process. For example, what is the most suitable strategy for implementation will depend on the specific needs of the target population and the specific context of the implementation setting [38, 64]. Using TMFs in this way can support the early identification of such needs and contextual preconditions [65]. For example, healthcare professionals working in teams may benefit more from strategies tailored to collaborative environments [66].

For successful implementation, it is important to address personal characteristics of the target group. For example, the GPs in this review assumed that older age is a barrier to using DHI. Interestingly, a study by Hermanus [67] showed that older people have a higher behavioural intention to use DHI. Nevertheless, the authors of that study also noted that older people use DHI less than younger people, which could be explained by the lack of digital skills among older people. Similarly, the National Institute for Public Health and the Environment (RIVM) confirmed that digital divide among Dutch elderly remains a concern [68]. These findings suggest that implementation strategies should focus not

on excluding older adults, but on strengthening digital literacy and providing tailored support to facilitate DHI uptake. At the same time, healthcare professionals recognize the benefits of DHIs, especially in its potential to personalize patient care [45, 46, 52]. They emphasize that in-person care remains essential for maintaining the patient-doctor relationship [51, 52, 69], but suggest that DHI could actually strengthen this patient-doctor relation by creating a sense of accessibility [51]. This further supports the value of integrating DHI into traditional care. This seems to particularly hold true for individuals with lower digital literacy. A blended care approach ensures that individuals who struggle with digital tools still receive adequate support while benefiting from the advantages of DHI [70].

Next, organisational considerations must be taken into consideration for maintaining patient engagement in self-monitoring DHIs, which remains a challenging part of implementation [71]. Patient experiences with DHI were mixed, as they reflected both potential benefits and challenges. While self-monitoring data can empower individuals, it can also be perceived as burdensome [72] and leading to negative emotional responses to glucose fluctuations [73]. This raises broader concerns about the psychological impact of continuous monitoring. For some, the belief that their healthcare professionals monitors their data serves as a motivating factor for adherence [41]. However, this reliance also raises ethical and legal questions regarding professional liability and the responsibility to intervene in cases of abnormal readings [51]. These complexities highlight the need for clear guidelines defining roles, responsibilities [41] and data security [40, 41].

In addition to that, from an organisational perspective, DHIs must integrate seamlessly with existing healthcare systems [51, 52] to enable sustainable implementation. This includes for example, embedding DHIs within electronic health records, aligning them with routine clinical workflows, and addressing reimbursement mechanisms. Successfully integrating blended care into routine practice can improve quality of care and contribute to a more personalized care system.

To overcome some of the identified implementation barriers and harness facilitators, it is highly advisable to invest in in-person training for healthcare professionals, ideally delivered before and during initial implementation with follow-up sessions as needed [38, 45, 64]. However, the effectiveness of training depends also on factors such as content, duration and facilitation of the training [74] and its attainability may be hindered by its associated costs [40]. A more structured approach to organizing efficient training, tailored to the specific needs of both professionals and patients, can support and improve sustainability of a blended care process.

In summary, this scoping review provides guidance for structured thinking about and concrete steps to take to optimize implementation of lifestyle related blended care for patients with obesity, overweight and/or T2DM. In considering concrete steps to take, using TMFs can support researchers and developers in making sure they address all relevant topics for implementation and enables cross-comparisons between implementation factors and processes in different projects. Furthermore, reducing organisational burden, ensuring usability through user-centred design, and embedding blended care within routine workflows and regulatory structures, together can strengthen long-term adoption.

Limitations

As with any scientific study, this study faced some limitations. The main one being the relatively low agreement between the first and second screeners for the initial 10% of the data. This discrepancy was attributable to the initially unclear interpretation of blended care and was resolved after discussions that led to the extended definition of blended care as described in the methods section. In addition, given the explorative nature of this paper, the authors placed emphasis on scoping the breadth of this field of study, rather than evaluating the quality of its data. In line with that, another potential limitation of this scoping review is its reliance on a narrative summary, which inherently involves subjective interpretations. Efforts were made to reduce bias by using a structured extraction form and regularly discussing interpretations with the second reviewer. Finally, all included articles unintentionally focused exclusively on primary care settings. This may limit the generalizability of the findings to other healthcare settings, such as secondary or tertiary care. Therefore, further research into other healthcare settings is needed.

Conclusion

Blended care and digital health interventions offer opportunities to enhance personalisation, patient involvement, and continuity of care. However, challenges such as high implementation costs, workflow integration issues, and the digital divide highlight that sustainable implementation requires addressing overarching domains rather than isolated factors.

At the same time, there is a notable gap in the literature on real world implementation, with most studies focusing on effectiveness. This underscores the importance of developing strategies that support adoption and sustainability, while ensuring uniform terminology and understanding of blended care and DHIs. Healthcare organisations can reduce organisational burden, policy-makers can support regulatory alignment, developers can

prioritize user-centred design, and healthcare professionals can integrate DHI into routine practice and support patients in its use. Ultimately, successful integration of DHIs depends on critical evaluation and, where necessary, redesign of care workflows to ensure DHI are meaningfully embedded in blended care.

Abbreviations

CFIR	Consolidated Framework for Implementation Research
DHI	Digital Health Intervention
GP	General Practitioner
NASSS	Non-adoption, abandonment, scale-up, spread and sustainability
NPT	Normalization Process Theory
RE-AIM	Reach, Effectiveness, Adoption, Implementation, Maintenance
T2DM	Type 2 Diabetes Mellitus

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s43058-026-00904-3>.

Additional file 1.
Additional file 2.
Additional file 3.
Additional file 4.

Acknowledgements

The authors utilized OpenAI's ChatGPT 4.0 (2024) for language verification, sentence structure suggestions, and overall readability enhancement.

Authors' contributions

All authors (C.L., A.M., G.D.L., and N.B.) contributed to the development of the study. In collaboration with an information specialist from the University of Twente, all authors formulated the search query. Data collection was conducted by C.L. and N.B., while C.L. was responsible for data extraction. C.L. drafted the initial manuscript, after which all authors (C.L., A.M., G.D.L., and N.B.) provided critical revisions and contributed to its finalization. All authors (C.L., A.M., G.D.L., and N.B.) read and approved the published version of the manuscript.

Funding

This project was funded by the Dutch Diabetes Research Foundation, project number 2022.30.001. The funder was not involved in the design, collection, analysis, or interpretation of the data, in the writing of the manuscript, or in the decision to publish this manuscript.

Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare that they have no competing interests.

Author details

¹Department of Biomedical Signals and Systems, University of Twente, Enschede, The Netherlands. ²Department of Internal Medicine/Nephrology, Ziekenhuisgroep Twente, Almelo, The Netherlands. ³Section of Psychology, Health and Technology, University of Twente, Enschede, The Netherlands.

Received: 14 August 2025 Accepted: 9 March 2026

Published online: 19 March 2026

References

- Noncommunicable diseases. Available from: <https://www.who.int/news-room/fact-sheets/detail/noncommunicable-diseases>. Accessed 23 Feb 2026.
- Sun H, Saeedi P, Karuranga S, Pinkepank M, Ogurtsova K, Duncan BB, et al. IDF diabetes atlas: global, regional and country-level diabetes prevalence estimates for 2021 and projections for 2045. *Diabetes Res Clin Pract*. 2022;1:109119. <https://doi.org/10.1016/j.diabres.2021.109119>.
- IDF Diabetes Atlas | Global Diabetes Data & Statistics. Available from: <https://diabetesatlas.org/>. Accessed 23 Feb 2026.
- Statistiek CB voor de. Meer obesitas en diabetes bij volwassenen met armoederisico. Cent Bur Voor Stat. 2022. Available from: <https://www.cbs.nl/nl-nl/nieuws/2022/40/meer-obesitas-en-diabetes-bij-volwassenen-met-armoederisico>. Accessed 23 Feb 2026.
- Hecker J, Freijer K, Hilgsmann M, Evers SMAA. Burden of disease study of overweight and obesity; the societal impact in terms of cost-of-illness and health-related quality of life. *BMC Public Health*. 2022;7(1):46. <https://doi.org/10.1186/s12889-021-12449-2>.
- Chen L, Pei J-H, Kuang J, Chen H-M, Chen Z, Li Z-W, et al. Effect of lifestyle intervention in patients with type 2 diabetes: a meta-analysis. *Metabolism*. 2015;64(2):338–47. PMID:25467842.
- Knowler WC, Barrett-Connor E, Fowler SE, Hamman RF, Lachin JM, Walker EA, Nathan DM, Diabetes Prevention Program Research Group. Reduction in the incidence of type 2 diabetes with lifestyle intervention or metformin. *N Engl J Med* 2002 Feb 7;346(6):393–403. PMID:11832527.
- Kelly MP, Barker M. Why is changing health-related behaviour so difficult? *Public Health*. 2016;136:109–16. PMID:27184821.
- Vollenbroek-Hutten M, Goolkate T, Lankheet M, Hegeman JH (Han), Laverman GD, Beijum B-J van. Insufficient Behavioral Change Skill Hampers Adoption of Ehealth Services. 2026. p. 119–125. Available from: <https://www.scitepress.org/Link.aspx?doi=10.5220/0006675501190125>. Accessed 23 Feb 2026.
- Öberg U, Isaksson U, Jutterström L, Orre CJ, Hörnsten Å. Perceptions of Persons With Type 2 Diabetes Treated in Swedish Primary Health Care: Qualitative Study on Using eHealth Services for Self-Management Support. *JMIR Diabetes*. 2018;3(1):e7. PMID:30291075.
- Huygens MWJ, Vermeulen J, Swinkels ICS, Friele RD, van Schayck OCP, de Witte LP. Expectations and needs of patients with a chronic disease toward self-management and eHealth for self-management purposes. *BMC Health Serv Res*. 2016;8(16):232. PMID:27391471.
- Yang Y, Lee EY, Kim H-S, Lee S-H, Yoon K-H, Cho J-H. Effect of a Mobile Phone-Based Glucose-Monitoring and Feedback System for Type 2 Diabetes Management in Multiple Primary Care Clinic Settings: Cluster Randomized Controlled Trial. *JMIR MHealth UHealth*. 2020;8(2):e16266. PMID:32130172.
- Talboom-Kamp EPWA, Verdijk NA, Kasteleyn MJ, Numans ME, Chavannes NH. From chronic disease management to person-centered eHealth; a review on the necessity for blended care. *Clin eHealth*. 2018;1(1):3–7. <https://doi.org/10.1016/j.ceh.2018.01.001>.
- Dennis SM, Harris M, Lloyd J, Powell Davies G, Faruqi N, Zwar N. Do people with existing chronic conditions benefit from telephone coaching? A rapid review. *Aust Health Rev Publ Aust Hosp Assoc*. 2013;37(3):381–8. PMID:23701944.
- Wagner EH, Austin BT, Davis C, Hindmarsh M, Schaefer J, Bonomi A. Improving chronic illness care: translating evidence into action. *Health Aff (Millwood)* Project Hope. 2001;20(6):64–78. PMID:11816692.
- Witkamp R, Navis G, Boer J, Plat J, Assendelft P, Vries J de, et al. Kennissynthese voeding als behandeling van chronische ziekten. Wageningen Universiteit; 2017; Available from: <https://research.wur.nl/en/publications/kennissynthese-voeding-als-behandeling-van-chronische-ziekten/>. Accessed 23 Feb 2026.
- Raber M, Liao Y, Rara A, Schembre SM, Krause KJ, Strong L, et al. A systematic review of the use of dietary self-monitoring in behavioural weight loss interventions: delivery, intensity and effectiveness. *Public Health Nutr*. 2021;24(17):5885–913. PMID:34412727.
- Singh B, Ahmed M, Staiano AE, Gough C, Petersen J, Vandelanotte C, et al. A systematic umbrella review and meta-meta-analysis of eHealth and mHealth interventions for improving lifestyle behaviours. *NPJ Digit Med*. 2024;7(1):179. PMID:38969775.
- Boels AM, Rutten G, Zuithoff N, de Wit A, Vos R. Effectiveness of diabetes self-management education via a smartphone application in insulin treated type 2 diabetes patients - design of a randomised controlled trial ('TRIGGER study'). *BMC Endocr Disord*. 2018;18(1):74. PMID:30348142.
- Vigersky RA. An overview of management issues in adult patients with type 2 diabetes mellitus. *J Diabetes Sci Technol*. 2011;5(2):245–50. PMID:21527089.
- Lyles CR, Harris LT, Le T, Flowers J, Tufano J, Britt D, et al. Qualitative evaluation of a mobile phone and web-based collaborative care intervention for patients with type 2 diabetes. *Diabetes Technol Ther*. 2011;13(5):563–9. PMID:21406018.
- Walthouwer MJL, Oenema A, Soetens K, Lechner L, de Vries H. Implementation of web-based interventions by Dutch occupational health centers. *Health Promot Int*. 2017;32(5):818–30. PMID:27020573.
- van Beurden SB, Simmons SJ, Tang JCH, Mewse AJ, Abraham C, Greaves CJ. Informing the development of online weight management interventions: a qualitative investigation of primary care patient perceptions. *BMC Obes*. 2018;5:7. PMID:29449953.
- Odnoletkova I, Buysse H, Nobels F, Goderis G, Aertgeerts B, Annemans L, et al. Patient and provider acceptance of telecoaching in type 2 diabetes: a mixed-method study embedded in a randomised clinical trial. *BMC Med Inform Decis Mak*. 2016;16(1):142. PMID:27825340.
- Hohberg V, Fuchs R, Gerber M, Künzler D, Paganini S, Faude O. Blended Care Interventions to Promote Physical Activity: A Systematic Review of Randomized Controlled Trials. *Sports Med - Open*. 2022;8(1):100. PMID:35907158.
- van der Kleij RMJJ, Kasteleyn MJ, Meijer E, Bonten TN, Houwink EJJ, Teichert M, et al. SERIES: eHealth in primary care. Part 1: Concepts, conditions and challenges. *Eur J Gen Pract*. 2019;25(4):179–189. PMID:31597502.
- Zhang M, Wang W, Li M, Sheng H, Zhai Y. Efficacy of Mobile Health Applications to Improve Physical Activity and Sedentary Behavior: A Systematic Review and Meta-Analysis for Physically Inactive Individuals. *Int J Environ Res Public Health*. 2022;19(8):4905. PMID:35457775.
- Romeo A, Edney S, Plotnikoff R, Curtis R, Ryan J, Sanders I, et al. Can Smartphone Apps Increase Physical Activity? Systematic Review and Meta-Analysis. *J Med Internet Res*. 2019;21(3):e12053. PMID:30888321.
- Villinger K, Wahl DR, Boeing H, Schupp HT, Renner B. The effectiveness of app-based mobile interventions on nutrition behaviours and nutrition-related health outcomes: a systematic review and meta-analysis. *Obes Rev*. 2019;20(10):1465–84. PMID:31353783.
- Wu Y, Yao X, Vespasiani G, Nicolucci A, Dong Y, Kwong J, et al. Mobile app-based interventions to support diabetes self-management: a systematic review of randomized controlled trials to identify functions associated with glycemic efficacy. *JMIR Mhealth Uhealth*. 2017;5(14):e35. PMID:28292740.
- Duan Y, Shang B, Liang W, Du G, Yang M, Rhodes RE. Effects of eHealth-based multiple health behavior change interventions on physical activity, healthy diet, and weight in people with noncommunicable diseases: systematic review and meta-analysis. *J Med Internet Res*. 2021;23(22):e23786. PMID:33616534.
- Wang J, Cai C, Padhye N, Orlander P, Zare M. A behavioral lifestyle intervention enhanced with multiple-behavior self-monitoring using mobile and connected tools for underserved individuals with type 2 diabetes and comorbid overweight or obesity: pilot comparative effectiveness trial. *JMIR Mhealth Uhealth*. 2018;6(10):e92. PMID:29636320.
- Geurts KAM, Woodcock-Nekeman S, Hummel M, Dietvorst CAW, van Rossum EFC, Berk KA. The effect of including eHealth in dietary interventions for patients with type 2 diabetes with overweight or obesity: a systematic review. *Nutrients*. 2023;15(29):3776. PMID:37686808.
- Greenhalgh T, Shaw S, Wherton J, Vijayaraghavan S, Morris J, Bhattacharya S, et al. Real-world implementation of video outpatient consultations at macro, meso, and micro levels: mixed-method study. *J Med Internet Res*. 2018;20(17):e150. PMID:29625956.

35. Ross J, Stevenson F, Lau R, Murray E. Factors that influence the implementation of e-health: a systematic review of systematic reviews (an update). *Implement Sci*. 2016;11(26):146. PMID:27782832.
36. Weik L, Fehring L, Mortsiefer A, Meister S. Understanding inherent influencing factors to digital health adoption in general practices through a mixed-methods analysis. *NPJ Digit Med*. 2024;7(27):47. PMID:38413767.
37. Berardi C, Antonini M, Jordan Z, Wechtler H, Paolucci F, Hinwood M. Barriers and facilitators to the implementation of digital technologies in mental health systems: a qualitative systematic review to inform a policy framework. *BMC Health Serv Res*. 2024;24(26):243. PMID:38408938.
38. Ross J, Stevenson F, Dack C, Pal K, May C, Michie S, et al. Developing an implementation strategy for a digital health intervention: an example in routine healthcare. *BMC Health Serv Res*. 2018;18(19):794. PMID:30340639.
39. Damschroder LJ, Reardon CM, Widerquist MAO, Lowery J. The updated consolidated framework for implementation research based on user feedback. *Implement Sci*. 2022;17(1):29. <https://doi.org/10.1186/s13012-022-01245-0>.
40. Ahmed B, Dannhauser T, Philip N. A systematic review of reviews to identify key research opportunities within the field of eHealth implementation. *J Telemed Telecare*. 2019;25(5):276–85. PMID:29703127.
41. Li J, Talaie-Khoei A, Seale H, Ray P, MacIntyre CR. Health Care Provider Adoption of eHealth: Systematic Literature Review. *Interact J Med Res*. 2013;2(1):e7. PMID:23608679.
42. Schreweis B, Pobiruchin M, Strotbaum V, Suleder J, Wiesner M, Bergh B. Barriers and Facilitators to the Implementation of eHealth Services: Systematic Literature Analysis. *J Med Internet Res*. 2019;21(11):e14197. PMID:31755869.
43. Tricco AC, Lillie E, Zarin W, O'Brien KK, Colquhoun H, Levac D, et al. PRISMA extension for scoping reviews (PRISMA-ScR): checklist and explanation. *Ann Intern Med*. 2018;169(7):2–473. PMID:30178033.
44. Covidence - Better systematic review management. Covidence. Available from: <https://www.covidence.org/>. Accessed 23 Feb 2026.
45. Schimmer R, Orre C, Öberg U, Danielsson K, Hörnsten Å. Digital Person-Centered Self-Management Support for People With Type 2 Diabetes: Qualitative Study Exploring Design Challenges. *JMIR Diabetes*. 2019;4(3):e10702. PMID:31538941.
46. Timm L, Karlsson I, Sidney Annerstedt K, Absetz P, Forsberg BC, Dai-vadanam M, et al. Intervention Fidelity Focusing on Interaction between Participants and Facilitators in a Telephone-Delivered Health Coaching Intervention for the Prevention and Management of Type 2 Diabetes. *Nutrients*. 2021;13(11):3862. PMID:34836116.
47. Sánchez A, Silvestre C, Campo N, Grandes G. Type-2 diabetes primary prevention program implemented in routine primary care: a process evaluation study. *Trials*. 2016;20(1):254. PMID:27206733.
48. Costa-Pinel B, Mestre-Miravet S, Barrio-Torrell F, Cabré-Vila J-J, Cos-Claramunt X, Aguilar-Sanz S, et al. Implementation of the DP-TRANSFERS project in Catalonia: a translational method to improve diabetes screening and prevention in primary care. *PLoS ONE*. 2018;13(3):e0194005. PMID:29543842.
49. Hanley J, Fairbrother P, McCloughan L, Pagliari C, Paterson M, Pinnock H, et al. Qualitative study of telemonitoring of blood glucose and blood pressure in type 2 diabetes. *BMJ Open*. 2015;5(12):e008896. PMID:26700275.
50. Brown JD, Hales S, Evans TE, Turner T, Sword DO, O'Neil PM, et al. Description, utilisation and results from a telehealth primary care weight management intervention for adults with obesity in South Carolina. *J Telemed Telecare*. 2020;26(1–2):28–35. PMID:30045661.
51. Ayre J, Bonner C, Bramwell S, McClelland S, Jayaballa R, Maberly G, et al. Factors for Supporting Primary Care Physician Engagement With Patient Apps for Type 2 Diabetes Self-Management That Link to Primary Care: Interview Study. *JMIR MHealth UHealth*. 2019;7(1):e11885. PMID:30664468.
52. Vonken L, Hussein H, Crutzen R, Vluggen S. Perceptions of Dutch general practitioners towards eHealth for patients with type-2 diabetes: a qualitative study. *Fam Pract*. 2023;40(1):9–97. PMID:35715556.
53. Suhrie L, Ayyagari R, Mba C, Olsson K, Torres-Aparcana H, James S, et al. The effectiveness of telemedicine in the prevention of type 2 diabetes mellitus: a systematic review and meta-analysis of interventions. *Diabetes Metab Syndr*. 2025;19(5):103252. PMID:40527161.
54. Xue H, Zhang L, Shi Y, Zhang H, Zhang C, Liu Y, et al. The effectiveness of digital health intervention on glycemic control and physical activity in patients with type 2 diabetes: a systematic review and meta-analysis. *Front Digit Health*. 2025;7:1630588. PMID:40799285.
55. Xiao Y, Wang Z, Zhang L, Xie N, Chen F, Song Z, et al. Effectiveness of Digital Diabetes Management Technology on Blood Glucose in Patients With Type 2 Diabetes at Home: Systematic Review and Meta-Analysis. *J Med Internet Res*. 2025;3(27):e66441. PMID:40053775.
56. Dat TV, Binh V, Hoang TM, Tu VL, Luyen PD, Anh LTK. The effectiveness of telemedicine in the management of type 2 diabetes: A systematic review. *SAGE Open Med*. 2024;12:20503121241271850. PMID:39263639.
57. Skivington K, Matthews L, Simpson SA, Craig P, Baird J, Blazeby JM, et al. A new framework for developing and evaluating complex interventions: update of Medical Research Council guidance. *BMJ Clinical research ed*. 2021;30:n2061. PMID:34593508.
58. Lee EW, McCloud RF, Viswanath K. Designing Effective eHealth Interventions for Underserved Groups: Five Lessons From a Decade of eHealth Intervention Design and Deployment. *J Med Internet Res*. 2022;24(1):e25419. PMID:34994700.
59. Greenhalgh T, Wherton J, Papoutsis C, Lynch J, Hughes G, A'Court C, et al. Beyond Adoption: A New Framework for Theorizing and Evaluating Nonadoption, Abandonment, and Challenges to the Scale-Up, Spread, and Sustainability of Health and Care Technologies. *J Med Internet Res*. 2017;19(11):e367. PMID:29092808.
60. RE-AIM – Home – Reach Effectiveness Adoption Implementation Maintenance. Available from: <https://re-aim.org/>. Accessed 23 Feb 2026.
61. Murray E, Treweek S, Pope C, MacFarlane A, Ballini L, Dowrick C, et al. Normalisation process theory: a framework for developing, evaluating and implementing complex interventions. *BMC Med*. 2010;20(8):63. PMID:20961442.
62. Nilsen P. Making sense of implementation theories, models and frameworks. *Implement Sci IS*. 2015;21(10):53. PMID:25895742.
63. Pong C, Tseng RMWW, Tham YC, Lum E. Current Implementation of Digital Health in Chronic Disease Management: Scoping Review. *J Med Internet Res*. 2024;12(26):e53576. PMID:39666972.
64. Kirkland EB, Johnson E, Bays C, Marsden J, Verdin R, Ford D, et al. Diabetes Remote Monitoring Program Implementation: A Mixed Methods Analysis of Delivery Strategies. Barriers and Facilitators Telemed Rep. 2023;4(1):30–43. PMID:36950477.
65. eHealth Research Theory and Development: A Multidisciplinary Approach. Routledge CRC Press. Available from: <https://www.routledge.com/eHealth-Research-Theory-and-Development-A-Multidisciplinary-Approach/Kip-Beerlage-deJong-vanGemert-Pijnen-Sanderma-Kelders/p/book/9781032295237>. Accessed 23 Feb 2026.
66. Cassidy CE, Harrison MB, Godfrey C, Nincic V, Khan PA, Oakley P, et al. Use and effects of implementation strategies for practice guidelines in nursing: a systematic review. *Implement Sci IS*. 2021;16(1):102. PMID:34863220.
67. van de Werken HA, Rohrbach PJ, Bolman CAW. Explaining intention and use of Mhealth with the unified theory of acceptance and use of technology. *Acta Psychol (Amst)*. 2025;254:104819. PMID:39985833.
68. Digitale zorg | RIVM. Available from: <https://www.rivm.nl/digalezorg>. Accessed 23 Feb 2026.
69. Lie SS, Karlén B, Graue M, Oftedal B. The influence of an eHealth intervention for adults with type 2 diabetes on the patient-nurse relationship: a qualitative study. *Scand J Caring Sci*. 2019;33(3):741–9. PMID:30866066.
70. Van Tuyl L, Keuper J, Standaar L, Alblas E, Keij B. Toegankelijkheid van digitale zorg: signalen uit de e-healthmonitor 2023. *TSG Tijdschr Voor Gezondheidswet*. 2025;103(S1):3–7. <https://doi.org/10.1007/s12508-025-00451-7>.
71. Patel ML, Wakayama LN, Bennett GG. Self-monitoring via digital health in weight loss interventions: a systematic review among adults with overweight or obesity. *Obesity*. 2021;29(3):478–99. PMID:33624440.
72. Torbjørnsen A, Ribu L, Rønnevig M, Grøttland A, Helseth S. Users' acceptability of a mobile application for persons with type 2 diabetes: a qualitative study. *BMC Health Serv Res*. 2019;6(1):641. PMID:31492176.

73. Perceived needs for supported self-management of type 2 diabetes: a qualitative investigation of the potential for a web-based intervention - UQ eSpace. Available from: <https://espace.library.uq.edu.au/view/UQ:325934>. Accessed 23 Feb 2026.
74. Varsi C, Solberg Nes L, Kristjansdottir OB, Kelders SM, Stenberg U, Zangi HA, et al. Implementation Strategies to Enhance the Implementation of eHealth Programs for Patients With Chronic Illnesses: Realist Systematic Review. *J Med Internet Res*. 2019;21(9):e14255. PMID:31573934.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.